

INFECTION CONTROL

NCLEX STUDY GUIDE



Organized for study and printing | 20 content pages

Infection Control

Chain of Infection control (organizer)

1. **Infectious agent** or pathogen
 - Weird name (bacterium, virus, fungus)
2. **Reservoir** or source of pathogen growth
 - Where the agent grows (ex: garapata en lyme disease)
3. **Portal of exit**
 - Cough, sneezing, saliva
4. **Mode of transmission** (part where the nurse work)
 - Here you take **precaution** to stop the chain
 - Standard - Airborne
 - Droplet - Contact
5. **Portal of entry**
 - Cuts, eyes, mouth, nose
6. **Susceptible host**
 - (Hombre) Next person sick.

Medical Asepsis or Clean technique (no sterile)	Surgical Asepsis (Sterile)
1. Hand washing. 2. Gloves, PPE 3. Hands are rinsed with finger pointing down. - Oral medication - Foley - Procedure in nose, mouth, vagina, and rectum Las manos se pueden limpiar con el hand sanitizer, except si esta sucias que se usa agua y javon, en el caso de C.Diff obligado siempre agua y javon	1. Scrub: Hands are rinsed with finger pointing upward above elbow. Principles of surgical Asepsis 1. All items used withing a sterile field must be sterile. 2. One inch (2.5cm) Margin around the edge are not sterile. 3. Wet surfaces = (Nonsterile) contaminate the sterile field. Board: If sterile field is contaminated with sterile water = it is not sterile field 4. Wound is sterile only from waist to shoulder. 5. Don't turn your back to the sterile field. 6. Open first away fold of the sterile kit, then laterals and finally the one close to sterile field (to me). 7. Hold Forceps with tips below the handle.

Precautions

Standard precautions = Universal or blood born

Δ All Patients

- Δ All body fluids **except sweat** are infectious
- Δ Mucosa and broken skin are infectious.
- Δ Practice hand hygiene before and after touching patients

Board: Disseminate herpes Soster = Multiple precautions! All precautions except **Droplets**

Note: (Only Standards Precaution)

1. HIV
 - Cryptococcal neoformans infection
 - Jyrovency Pneumonia carinii Pneumonia
 - Kaposi's sarcoma
2. Hepatitis
 - A, B, C, D
 - A is contact if diarrhea or bowel incontinence
3. Mononucleosis infection (beso)
4. Anthrax
5. Kawasaki = vasculopathy
6. Legionary Disease = pneumonias de spores y hot water.
7. Histoplasmosis = (cacas de aves) 8. Aspergillosis
8. **Aspergillosis**
9. Sarcoidosis
10. Osteomyelitis (Bone Infection)
11. Glomerulonephritis
12. Endocarditis

Personal Protective Equipment (PPE)

Wear/Don/Donning	Removing
1. Gown: first neck and then waist 2. Mask 3. Goggle or eye shield 4. Gloves	4. Gloves 3. Goggles 1. Gown 2. mask

1. Mask

N-95 =

- Heppa-filter
- particulate mask
- special mask
- For Pt in airborne precautions for nurse or family member when going inside the room in negative pressure, before entering in the room.
- Remove the mask when are outside the room and the door is close

Airborne disease?

Measles: if you have before the disease or get vaccinated don't need any mask bc you are immune

Chicken Pox: if you have before the disease or get vaccinated don't need any mask bc you are immune

TB: all nurse entered in the room need to wear N-95

Other that are airborne bc are multiple precaution.

SARS only N-95 when are in negative pressure. MERS

Regular or surgical mask use when:

- Nurse who going to see a Pt in droplet precaution only if less than 3 feet from Pt
- Pt in airborne and droplet when going outside the room.

Board: Pt with Tb and 3 sputum negativos que precaution tiene que seguir

Δ Usar clean gloves cuando cambia las sábanas de la cama.

2. Gown:

Always in contact precautions with gloves.

When risk for splash

- Wound care
- wound irrigation
- Tracheostomy care
- Bath

3. Gloves

Always in contact precaution

When contact with mucosa or body secretions

4. Eye shields

Only use in risk for splash

Needle precaution.

Board Sata: Never recap

Or Use one hand technique to recap.

○ Needle container or puncture resistance container discard when $\frac{3}{4}$ full.

Board: If full and no body empty, nurse have to empty it.

At home a bottle milk with water and soap = OK

○ If needle injury squeezing the finger or area with water (en la pila de agua)

- Lines = transport in a leak proof gab
- Body fluid spills: Dilute with bleach solution 10:1

1. Infectious Mononucleosis (Kissing disease)

Agent: Epstein bar virus

- Incubation period: 4-6 weeks
- Source: Oral secretion = Saliva

Assessment

1. Teenager
2. Fatigue = Persistent 1-2 years after
3. Sore throat, Fever, malaise, headache, nausea, abdominal pain.
4. Lymphadenopathy
5. Skin rash
6. 15-25 years old.

Board: if a teen or young adult has fatigue or sore throat = Infectious mononucleosis.

If an adult has the same symptom = Amygdalates.

Intervention

1. Standard precaution
 - Hand washing
 - Do not share the food or utensils
 - Do not kiss
2. Check for sign of splenic rupture

- Board like abdomen pain radiates to the left shoulder.

Note: Todas las rupturas de vísceras tienen el mismo síntoma = board like abdomen + pain to the left shoulder

Rocky mountain spotted fever (standard precaution)

Agent: Rickettsia Rockettsi

- Source: Tick rodents and dogs
- Incubation period = 2-14 days
- Transmission: Bite of infected tick (Dog)

Nota: si la garapata viene del deer (venado)= Lyme disease

Assessment

1. Fever, malaise, vomiting, headache, myalgias.
2. Maculopapular rash or petechiae in Ankles and wrists more frequent in palms and soles

Intervention

- Teaching = Same to Lyme disease
- 1. Wear long sleeves shirts, long pants tucked into long socks, pulled up over the leg part.
 - ☐ Light colors
 - ☐ Wear a hat.

2. Following paths (trillos) rather than walking in tall grass and shrub areas.
3. Apply insects' repellent before possible exposure to areas where Thicks are found.
4. Keeping yards at home trimmed and free of accumulated grass.
5. Apply repellent to dogs.
6. Saving the thick for later identification or discarding in toilet.
7. Use twister to remove it.

Board: Follow up? = a person who squeezing the thick. (Has to be send it to the lab or flushing in toilet)

Lyme Disease = Erythema Migrants (standard)



Agent: *Borrelia burgdorferi*

Transmission: Tick bite = Deer (venado)

Assessment

- Erythema Migrants
- Bull eye lesion or target lesion.

Stage 1	Stage 2	Stage 3
- Flu like symptoms - Skin lesion: Circular expanding rash with target like or bull's eye or ring lesion	- Cardiac problems - Bell's palsy (nerve VII) parte de la cara paralizada (cardiac + neuro)	- Arthritis - Chronic fatigue (Can last for years)

Treatment

1. Remove the thick as soon as possible with tweezers (never scraping)
- Flushing in the toilet or sealed in a jar and send it to the hospital.

Note: este es lo contrario del bee Sting que se remove by scrapping.

2. Antibiotic

- ☐ Tetracycline,
- ☐ Doxycycline or
- ☐ Penicillin
- ☐ ASAP cuando sospechas que es Lyme D. sin tener la confirmation.
- 4- 6 weeks after, return to the hospital to confirm diagnostic (los mismos test del SIDA)
Δ ELISA

Δ WESTERN BLOT

➤ **TINEA CORPORIS** = Ring worm & tinea pedis = Athlete's foot (standard)



Board: Follow up? Question antibiotics order because it is a fungal infection.

Intervention /Prevention

- Use water shoes in public showers.
- Keep foot dry

Treatment

1. Ketoconazole (liver study before bc hepatotoxicity)
2. Miconazole
3. Clotrimazole

Side Effect

- Hepatotoxicity = Check liver function

CONTACT PRECAUTION

- Always use gown and gloves
- Use face shield or eye shield if risk for splash
- Dedicate equipment= Stethoscope, BP cuff = one for each patient or clean with alcohol between patients
- Do not use your personal equipment.
- Remove all PPE before going outside the room
- Door Open
- Can cohort with same infectious Agent.
- If patient goes outside – Inform the other department and transport what kind of precaution (contact, airborne, droplet) but not the agent (name of infection).
- If wound: Cover wounds for transportation.

Pathologies CONTACT Precaution

1. Diarrhea.
 - Gastroenteritis
 - Hep A if Diarrhea
 - C.Diff (long term antibiotic and prazoles)

- Rotavirus
- Enterovirus
- Shigella
- Salmonella
- 2. RSV
- 3. VRSA (vancomycin resistant staphylococcus)
- 4. MRSA (methicillin resistant staphylococcus)
- 5. Impetigo
- 6. Conjunctivitis
- 7. UTI E. Coli
- 8. Scabies
- 9. Pediculosis
- 10. Herpes Zoster (shingles)
- 11. Herpes Simple

MRSA & VRSA= Wound infection

If Disseminated Herpes Zoster = multiple precautions

- **Impetigo** (contact precaution)



- **Bacterial Infection**
 1. B hemolytic Streptococcus
 2. Staphylococcus infection

Note: Después de strep throat, scarlet fever or streptococcal impetigo puede desencadenar a glomerulonephritis or rheumatic fever

Ex: impetigo with a new murmur! = Call Dr bc complication with rheumatic fever

If you have impetigo and now have edema around de eyes call Dr bc complication with a glomerulonephritis)

Assessment

1. Occur due to poor hygiene
2. The most common site of infection is in the face around the mouth or nose.
3. The lesions begin as a vesicle or pustules surrounded by edema and redness.
4. - The lesion progress to exudative and crusting stages, and vesicular fluid becomes cloudy leaving Honey Crusted lesion.
5. Pruritus
6. Burning
7. Enlarged lympho-nodes.

Intervention

1. Contact Isolation until 24-48 hours after antibiotic = Bactrim Ointment.
 2. Allow lesion to dry by air exposure
 3. Assist the child with daily bath and antibacterial soap.
 4. Apply warm compress (normal saline) 2-3 times per day.
- Followed by a mild soap and water to remove the crust.
 - Burrow's solution may be used (to remove crust)
 - Then apply antibiotics. (Bactrim)
 - 5. Use emollients to prevent skin cracking
 - 6. Separate linens, clothes, dishes, etc.

➤ Pediculosis (contact) (piojos)

1. Infestation of the hair and scalp with lice
2. Incubation period 7- 10 days
3. Lice can survive 48 hours away from the host
4. Lice communication- cannot go to the school
5. Nits (liendra) no communication, child can go to the school.

Board: Teaching family to save in plastic bag non washable stuff or no using.

Intervention

1. Permethrin 1% cream rinse (nix) or shampoo que viene para eso en kit.
 2. Daily removal of nits with extra- fine tooth metal nit comb. Use gloves.
 3. Hairbrush or combs should be discard or soak in boiling water for 10 minutes.
 4. Instruct parents that non-essential bedding, toys, and clothing can be store in a tightly seal bag for 2 weeks (15 days). Board.
 5. Furniture and carpets need to be vacuumed
- Nota: los perros no cogen piojos.

➤ SCABIES (Contact)

- Parasite skin disorder (keyword = Burrow lesion)

Assessment

1. Pruritic rash, more frequently at night
2. Sites more affected
 - Axillar
 - Ano-genital area
 - Wrist
 - Hands (webs, between finger)
3. Burrow lesion fine grayish red lines

Intervention

1. Topical application of permethrin
2. Apply after bath or at night and remove in the morning 10-12h after (10-12h).
3. Lindane = Apply below the neck
4. Contraindicate in children less than 2 years and pregnant woman (bc teratogenic) →to prevent Seizure.

5. All clothes, bedding and pillowcases need to be changed daily.
6. Washed in hot water and dried in a hot cycle.
7. Non washable toys and other items should be sealed in plastic bag **for 1 week.**
8. **Repeat Tx 1 week after.**

Droplet Precaution

- Only use surgical mask or regular mask if less than 3 feet from the patient.
- Patients wear a mask when going out of the room.
- Private room or cohort with same infection
- **Door open (board Nurse left door close = Negligence)**

Pathologies **DROPLET** Precaution



S	Sepsis/2-scarlet fever/3-streptococcal pharyngitis or strep throat
P	Parvovirus B19 (5 th diseases)/ 2- Pertussis (whooping cough)/ 3-Pneumonias- 1- Pneumococcal or streptococcal pneumonia 2. Haemophiles influenzae pneumonia 3. walking pneumonia(atypical) 4. Plaque pneumonic (bioterrorism)
I	Influenza
D	Diphtheria
E	Epiglottitis
R	Rubella (bella german measles)
M	Mumps/ Bacterial Meningitis
A	Adenovirus
N	Norovirus
	Laryngotracheobronchitis (CROUP) (casi siempre provocado por influenza)

Rubella – German Measles (Droplet)

- Highly Teratogenic (if rubella in pregnant, baby born with malformation).
- All Pregnant woman – **Rubella titer (20-28 weeks)**

If less than 1:8= Administer rubella vaccine to mother after giving birth – within 72 hours after discharge

To avoid malformation to the next fetus

Wait 3 months to get pregnant again

If titer = 1:10 or more = ok, no action in needed

Agent: Rubella virus (no treatment bc virus)

Incubation period: 14-21 days

Communicable period: 7 days before and 7 days after rash.

Board: Child with rubella when send him to the school? = 7 days after rash

Assessment

1. Low grade fever
2. Malaise
3. Pinkish red maculopapular rash begins on the face and spreads to entire body within 1-3 days after.
4. Generalize lymphonodes adenopathy (most commonly suboccipital)
5. Occipital or suboccipital lymphonodes, (Adenopathy).

Intervention

1. Symptomatic = rest and fluids.
2. Isolate the infected child from pregnant woman (family, staff)

Board: mom had rubella during pregnant, can the newborn stay with mom, yes! (7 days after, no longer communicable.

Pertussis (whooping cough) (droplet)

Agent: Bordetella Pertussis (bacterial)

Incubation Period: 5 – 21 days

Communicable period: During catarrhal stage

Assessment

Catarrhal Stage	Paroxysmal (no communicable)	Recovery
- Cough - Mild upper respiratory symptoms. - Mild Fever - Conjunctivitis	- Cough episodes - Severe burst of cough with inspiration Whoop. - During this episode, child become flushed eyes, - Tongue protrusion that relieves when patient vomits	Back to normal

Treatment

1. Droplet Isolation 24 – 48 hours after antibiotic
2. Decreased environmental factors such as: dust, smoke, sudden change in temperature.
3. Suction and humidified O2.

Nota: Todo lo bacterial, 48 hours después del antibiotic deja de ser isolation, except TB.

Diphtheria (droplet)

Agent: Corynebacterium Diphtheria

Incubation period: 2-5 days

Communicable: variable

Keyword: Bull neck, Tracheostomy set at bed side = Priority

Assessment

1. Low grade fever
2. Malaise

3. Sore throat
4. Foul smelling mucopurulent nasal discharge.
5. Dense pseudo membrane of the throat
6. Lymphadenitis – neck edema = Bull neck

Treatment

1. Isolation for 24 – 48 hours after Isolation
2. Administer Diphtheria antitoxin
3. Bed Rest
4. Administer antibiotic
5. Tracheostomy at bed side

Scarlet Fever

Keyword = Strawberry tongue

Sandpaper rash

Pasties line

- ☐ Acute infection disease
- ☐ Incubation period: 2-7days
- ☐ Communicable period: variable (+- = 2 weeks after) Cause
- B nemolytic Streptococo (mismo de amigdalitis e impétigo, strepthroat)

Complications (OJO)

- Glomerulonephritis = Edema around de eyes
- Rheumatic fever= New murmur → Endometritis
- Call the Dr.

Assessment

1. Abrupt high fever, vomiting, headache, enlarged, lymphonodes, malaise, abdominal pain.
2. A red fine, sandpaper like rash or sunburn rash (axilas., groin, and neck, then spread to cover entire boy, except face.
3. Rash blanches with pressure except in areas of deep creases and folds = pastias line.
4. Skin desquamation.
5. Strawberry tongue.

Treatment

1. Droplet precaution 24-48 hours after antibiotic
2. Bed rest.
3. Encourage fluid intake.

MUMPS (Droplet)

Agent: Paramyxovirus

Incubation period: 14 – 21 days

Communicable: Immediately before and after parotid gland swelling.

Nota: Inflamación de las gonadas puede causar infertility.

Board: Que se le pregunta a un adult que refiere infertilidad? Tuviste mump cuando eras niño?.

Assessment

1. Extreme pain when ingesting sour or acidic substances (cuando se te hace agua la boca)
2. Jaw or ear pain aggravated by chewing
3. Parotid swelling
4. Fever
5. Headache
6. Orchitis may occur = can cause infertility

Complication

- Infertility
- Meningitis = Neck stiffness, fever = Call Dr.

Intervention (MUMPS)

1. Bed rest until testicular swelling subside
2. Use scrotal support
3. Use well snug underpan
4. Apply warm compress to relieve orchitis.
5. Apply hot or cold compress to the neck.

Erythema Infectiosum (Droplet)

□

Agent: Parvovirus B-19

Incubation: 4-14 days

Communicable: until rash appear (if you see the rash, it is no longer contagious).

Assessment

1. Erythema of the face lasting 1-4 days
2. 1 day after the rash appear on the face, spread to the extremities.
3. Low grade fever

Intervention

1. Children is usually not hospitalized
2. Avoid contact with pregnant woman
No pregnant RN, No pregnant family, No pregnant Staff.

Airborne Precaution

- Single room with negative pressure

- ☐ 6 – 12 exchange of air per hour
- ☐ Door Close
- ☐ Particulate, fitted Hepa filter mask = N95
- ☐ N-95 mask for nurse or family member inside the room
- ☐ TB always use N-95

Pathologies **AIRBRONE** Precaution

- Measles (rubeola) **Koplik's spots**
- Chicken pox (varicella) Smallpox
- TB
- For chicken pox and measles only use N-95 if not immune staff:
 - No had the disease before
 - Not had the vaccine before

Rubeola Measles (Airborne)

- ☐ **Acute viral disease**
- ☐ **Agent:** Paramyxovirus
- ☐ **Communicable:** 5 days before and 5 days after rash
- ☐ **Keyword:** **Koplik's Spots**

Assessment

1. 3Cs disease

- ☐ Coryza
- ☐ Cough
- ☐ Conjunctivitis

2. Erythema maculopapular starting on the face and spread downward to entire body.

- ☐ Turning brownish (after)
- ☐ Lasting 6-7 days

3. **Koplik's spots**

- ☐ Small red spots with bluish one center and a red base on the buccal mucosa.

Nota: N-95 if no immune staff

If vaccine or history of rubeola, no need for N-95

Same to chicken POX.

Intervention

1. Bed rest
2. Cool mist vaporizer (cough & sneezes)
3. Dim light if photophobia
4. **Antipyretic for fever Never ASA= to prevent Reye Syndrome**

Reye syndrome

Nota: La **ASA** se puede administrar en bacterial infection (eg: Kawaqsaky síndrome)
pero no in viral infection

Chickenpox = Varicela (airborne)

☐ **Agent:** Varicella Zoster virus

☐ **Incubation:** 14 – 21 days

☐ **Communicable:** 7 days before and 7 days after rash

- **Child or adult can return school/work when all lesion are crusted (dry)** Keyword: **At the same time occur different stages of types or healing lesion**

Board: picture with only one lesion = Impetigo

Several = Chicken pox

Foto con lesion en antebrazo = Ántrax

Assessment (chickenpox)

1. Headache, fever, malaise
2. **Rash in different stages of healing at the same time**
 - ☐ Papulas
 - ☐ Vesiculas
 - ☐ Pustulaes
 - ☐ Crust
- Rash appear on the trunk, scalp, vagina, rectum, genital area - **Except palm of hands and soles.** **Intervention (SATA)**
 1. Bath with calamine, Cetaphil, oatmeal, baking soda.
 2. Diphenhydramine for pruritus
 3. Apply mitten restrain to the child to prevent skin lesion infection
 4. **Never use ASA**

Nota: one RN solo para ellos (No RN pregnant bc immunocompromise)

Mitten restrain en todo lo que pique para que no se pueda rascar.

Herpes Zoster (shingles)

If Herpes Zoster = Contact Precaution

If Disseminated or generalized Herpes Zoster = **Multiple** precaution =
Airborne – Contact – Standard

Assessment

1. Pain (paresthesia)
 - ☐ Tingling, burning, numbness, piercing sensation along neuronal pathway or in a dermatomal pathway (recorrido del nervio).
2. **Allodinia** = increased stimuli sensitivity
3. Malaise
4. Fever
5. Headache

Treatment Management

1. Moist dressing to decreased pain = water, NSS or Burrow's solution.
2. Acyclovir during 1st 72 hours.
3. Carbamazepine or gabapentin for pain
4. Vaccination (Zostavax) – administer in patients > 60 years old.

Complication

1. Visual loss (board: patients with herpes simplex or zoster need ophthalmology consult)
2. Neurologic problems (meningitis)

Multiple Precaution**1. Disseminated or generalized Herpes Zoster**

- ☐ Airborne
- ☐ Contact
- ☐ Standard

2. MERS- Middle East respiratory Syndrome

- ☐
- ☐ Airborne
- ☐ Contact
- ☐ Standard

- Cause = Coronavirus = Severe respiratory Illness

- ☐ Fever
- ☐ SOB
- ☐ Cough
- ☐ Respiratory insufficiency

3. SARS = Cause = Coronavirus**- Severe acute respiratory syndrome**

- ☐ Fever
- ☐ SOB
- ☐ Cough
- ☐ Headache

- ☐ Respiratory Insufficiency

4. Ebola

- ☐ Hemorrhagic Fever
- ☐ Mosquito bite (agent) ☐ Multiple Precaution
 - ☐ Droplet
 - ☐ Contact
 - ☐ Standard
 - ☐ Airborne

Symptoms

- Petechiae
- Fever
- Hemorrhage
- Use condom 3 months after Ebola because transmission = body fluids

Biological Warfare Agents

1. Anthrax (standard)

- ☐ Cause by bacillus anthrax
- ☐ Transmitted by direct contact to the spore (NO person to person)

Assessment (Anthrax)

Inhaled	Skin (cutaneous)	GI
- Inhalation of spores - Flu like symptoms ☐ Fever ☐ Severe respiratory problems with hemoptysis	- Spores enter the skin through cuts and abrasion. - Itchy bump like mosquito bite. - Small liquid filled sac (ampollas) - Painless ulcer with black areas = Necrosis	- Following ingestion of contaminated undercooked food. ☐ Nausea, vomiting ☐ Abdominal pain ☐ Diarrhea: GI bleeding

Intervention

1. Decontamination period = Nurse use all PPE
 - ☐ Do not agitate clothes
 - ☐ Do not pull over
 - ☐ Cut clothes with scissors
 - ☐ Instruct patient to shower with shampoo, soap and water (el tiempo que quiera).

2. Decontaminated surface with bleach solution.

Treatment

Ciprofloxacin (choice) or Doxycycline

3. Botulism (standard)

- ☐ Bacterial Infection
- ☐ Contaminated food
 - ☐ Bulging or depressed canned food used
 - ☐ Ingestion of honey (NO honey in children less than 1 year)

Assessment (same to Myasthenia Gravis)

1. Descending paralysis
2. Ptosis (drooping eyes)
3. Weakened jaw clench
4. Disphagia
5. 12- 72 hours after exposure – Severe respiratory dysfunction and death.

Intervention

1. Antitoxin botulinic

Board: Cual de las bioterrorism lleva una toxina = Botulismo

2. Heating food (solido or liquid) 212° F at least 10 minutes to inactivated bacterial
3. Antibiotic = Aminoglycoside = Los que terminan en mycin y que no tenga THRO.

Nota: el botox es igual a un botulism

3. Plague yersinia Pestis

☐ Transmite through Fleabites (pulgas)

Bubonic Plague (standard)	Septicemic Plague (standard)	Pneumonic Plague (Droplet)
- Fever - Headache - Lymphonodes enlarge	- Generalized Infection - More severe	- Cough - Fever - Sputum - Pleuritis (chest pain)

Treatment

- Streptomycin or gentamycin

4. Smallpox = Chicken Pox (Airborne)

1. High fever, malaise, severe aching pain, flu like symptoms
2. Skin rash (more dense on the face)

Intervention = Chickenpox

1. Mitten restrain
2. Trim nails to prevent skin infection
3. Benadryl for pruritus

Board: cual de las de bioterrorism is not estándar precacution? =Smallpox = Airborne

5. Tularemia (standard)

□ Transmitted by rabbits and their ticks

- Fever - Muscle ache - Weakness
- Chills - Joint pain - Headache
- Pneumonia

Treatment (tularemia)

- Streptomycin
- Gentamycin 10-21 days
- Doxycycline

6. Mustard Gas

□ Chemical warfare agent that can cause severe chemical burn with:

- Pain free blister
- Difficulty breathing

Board: Cual es el gas que te quema? = Yellow or orange color with garlic or onion odor.

Assessment

1. 2nd or 3rd degree burn injuries
2. Severe eye damage or irritation corneal damage.
3. Mild to extreme respiratory distress

No antidote or treatment

7. Sarin Gas (Cholinergic)

□ Nerve Agent that cause severe disruption to the nervous system and interferes with muscle control and organ function.

(Te paraliza y mata en minutos a hours)

- Odorless
- Tasteless
- Colorless

Assessment

Cholinergic:

- Lacrimation
- ↑ Sweating
- ↑ Urinary output
- Seizure
- Death occur in few minutes to hours
- Salivation
- Diarrhea
- loss of consciousness
- Floppy paralysis

Treatment

- Antidote:
 - Atropine □
 - Pralidoxime.

CDC vaccination program

1	2	4	6	12 15 18	19-23 2 4-6 11-12 16
Birth	M	onths		Years	
Hep B	Hep B		Hep B		
#1	#2		#3		
RV	RV	RV*			
#1	#2	#3			
DTaP	DTaP	DTaP	DTaP	DTaP	
#1	#2	#3	#4	#5	
Hib	Hib	Hib	Hib		
#1	#2	#3	#4		
PCV	PCV	PCV	PCV		
#1	#2	#3	#4		
IPV	IPV		IPV	IPV	
#1	#2		#3	#4	
			Hep A		
			#1&3		
			MMR	MMR	
			#1	#2	
			VZV	VZV	
			#1	#2	
				Tdap	
				#1	
				HPV**	
				#1,2	
				MCV MCV	
				#1 #2	
			Inactiv	ated influenza annually	
				MenB	
				#1,2	
Hep B=Hepatitis B				IPV=Inactivated poliovirus	
RV=Rotavirus (2 doses if	Rotanix is used, 3 doses if RotaTeq is used)			MMR=Measles, mumps & rubella	
DTaP=Diphtheria, tetanus toxoids, acellular pertussis				VZV=Varicella-zoster virus	
Tdap=Tetanus toxoid, reduced diphtheria toxoid, acellular pertussis				Hep A=Hepatitis A	
Hib=Haemophilus influenzae type b conjugate vaccine				MCV=Meningococcal conjugate vaccine	
PCV=Pneumococcal conjugate vaccine				HPV*=Human papillomavirus, 3 doses if initiated age >15	
MenB=Meningococcal serogroup B vaccine					

QUWond